OPPORTUNITY BRIEF · SALES AND PRESALES

Post-quantum cryptography migration for defense systems to meet CNSA 2.0 compliance, covering cloud, enterprise, embedded, OT, tactical, and national-security systems.

Defense × Post-quantum cryptography migration × Post-quantum cryptography · OS016 v3 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
78 is the world moving	**53** can we play, can we win	**MEDIUM** 1 named in the evidence	**€95.9m** observed evidence · per year	**LATER** research and standards activi...	**L0** 20 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 0.0 published reference(s) in Defense, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

Post-quantum cryptography (PQC) migration for defense systems is a multi-year, multi-domain transformation required to meet NSA CNSA 2.0 compliance across cloud, enterprise, embedded, OT, tactical, and national-security systems. This opportunity helps defense organizations inventory cryptographic assets, plan migration, and implement quantum-safe solutions to protect against harvest-now-decrypt-later threats. The urgency comes from finalized NIST standards and the risk of adversaries recording encrypted traffic today for future decryption.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Observed: tendered contract value, annualised	€95.9m	€95.9m	€1.4m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

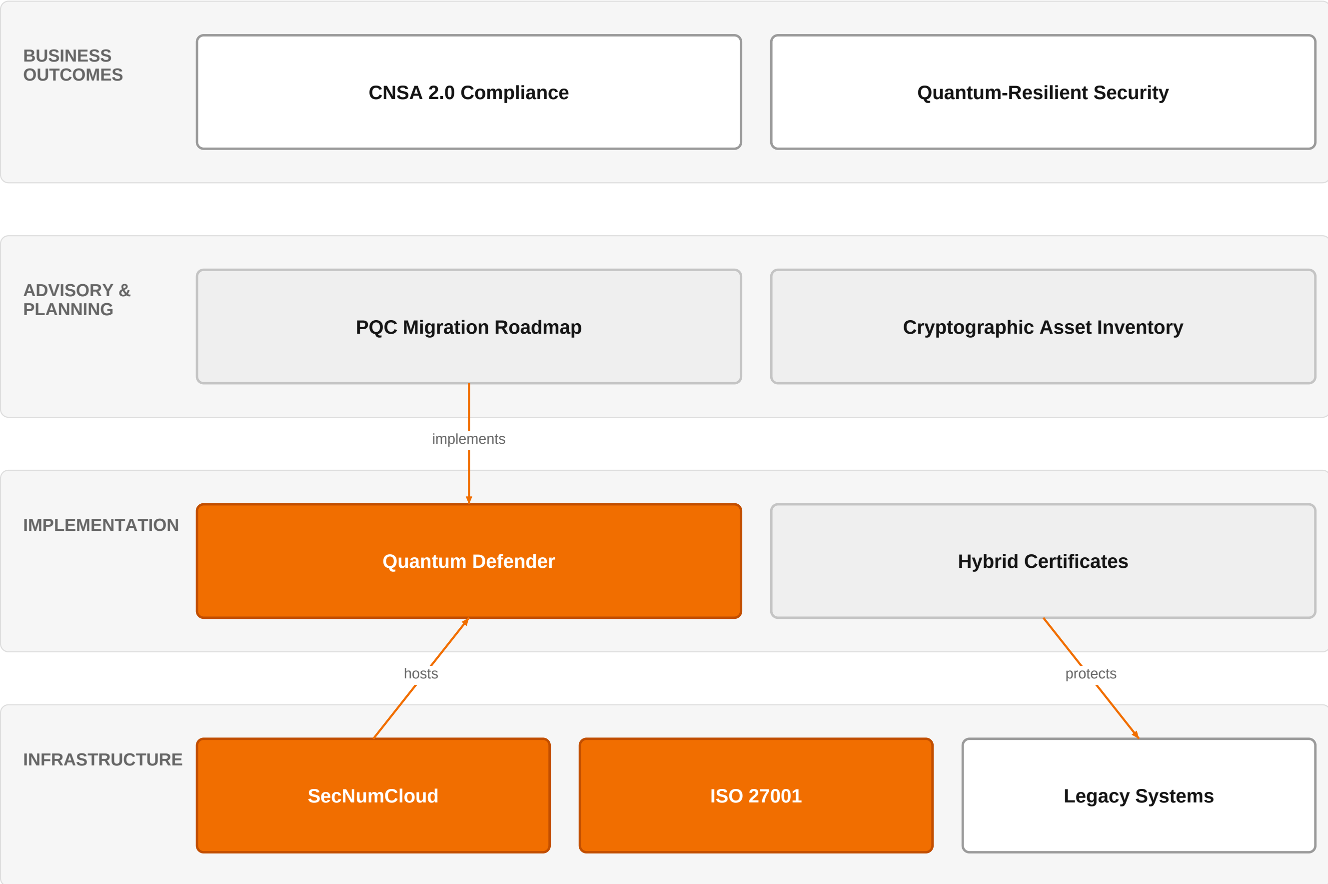
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€95.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-21..2026-08-05
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 5 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 3 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

The solution, in one picture

PQC Migration for Defense Systems



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's advisory and implementation services, built on sovereign cloud and security certifications, enable a defense customer to achieve CNSA 2.0 compliance and quantum-resilient security.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

NIST has finalized PQC standards (FIPS 203, 204, 205), initiating an unprecedented cryptographic migration, yet enterprise adoption remains critically low. The NSA's CNSA 2.0 mandates a multi-year, multi-domain transformation across cloud, enterprise, embedded, OT, tactical, and national-security systems. Adversaries can record encrypted traffic today and decrypt it later (harvest-now-decrypt-later), making the threat immediate even before a cryptographically relevant quantum computer exists. Additionally, space encryption hardware is getting a fivefold boost, indicating a shift toward more capable encryption solutions for space applications.

`Sources: SIG-B5F1E5C72CAB Springer Science and B 2026-07-30; SIG-57F4F0DAE044 Cryptography 2026-06-18; SIG-CA65D28B8AC6 openalex.org 2025-12-18; SIG-7ECDD0A30D62 techtimes.com 2026-08-13`

Who buys, and why now

The Chief Information Security Officer (CISO) or Chief Technology Officer (CTO) in defense organizations signs off on PQC migration, but the pain is felt by security architects and system engineers who must inventory and migrate cryptographic assets across diverse systems. The trigger is often a compliance deadline (e.g., CNSA 2.0) or a risk assessment highlighting harvest-now-decrypt-later exposure. Operational pain includes the complexity of migrating PKIs that serve millions of users, and the difficulty of discovering cryptographic assets in embedded systems that dominate defense platforms.

`Sources: SIG-C822609B6A07 openalex.org 2026-05-31; SIG-13AAE2CD702E MDPI AG 2026-01-19; SIG-CA65D28B8AC6 openalex.org 2025-12-18`

Why it is hot now — every claim bound to a dated source

- The advent of quantum computers poses a fundamental threat to current cryptography-based applications.
  [SIG-C822609B6A07]

- The migration of Public Key Infrastructures poses specific challenges, as these systems are complex, serving millions of users and fundamental in providing cybersecurity. [SIG-C822609B6A07]
- Telecommunication infrastructures rely on cryptographic protocols designed for long-term confidentiality, yet data exchanged today faces future exposure when adversaries acquire quantum or large-scale computational capabilities. [SIG-CA65D28B8AC6]
- Agentic AI systems depend on classical public-key cryptography for agent identity, tool invocation, inter-agent communication, model integrity, and persistent state, exposing them to a cryptographically relevant quantum computer. [SIG-81E56EEBF9D4]
- The impending threat posed by quantum-capable adversaries necessitates a secure and practical transition of Public Key Infrastructures (PKIs) to support post-quantum cryptography (PQC). [SIG-53F3508BD517]
- Hybrid certificate designs can provide quantum-resistant security. [SIG-53F3508BD517]
- Adversaries can record encrypted traffic today and decrypt it later, a tactic known as harvest-now-decrypt-later. [SIG-B8561992126B, SIG-CA65D28B8AC6]
- PQC migration to NIST FIPS 203, 204, and 205 under the NSA Commercial National Security Algorithm Suite (CNSA) 2.0 is a multi-year, multi-domain transformation across cloud, enterprise, embedded, operational technology (OT), tactical, and national-security systems. [SIG-57F4F0DAE044]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver a PQC migration assessment and roadmap, leveraging its Cybersecurity experts and Defense and security experts to conduct cryptographic asset discovery and inventory across the customer's systems. The engagement would start with a gap analysis against CNSA 2.0, followed by a prioritized migration plan covering cloud, enterprise, embedded, OT, and tactical domains. Orange would then implement quantum-safe solutions, potentially using its Quantum Defender / quantum-safe offers, and ensure compliance with ISO 27001 and SecNumCloud certifications where applicable.

Why Orange can win

Orange brings a unique combination of cybersecurity expertise, defense domain knowledge, and sovereign cloud certifications (SecNumCloud) that are critical for defense customers. The Quantum Defender / quantum-safe offers provide a concrete solution, and Orange Group researchers can contribute to cutting-edge PQC integration. However, the assets are somewhat thin; Orange does not have a dedicated PQC product suite like some competitors, so the win must be based on trust, sovereignty, and integration capability.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Quantum Defender / quantum-safe offers	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 6.3; 1 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include Microsoft (also a partner), Atos/Eviden, IBM, Airbus Protect, Entrust, ID Quantique, Thales, and Dassault Systemes/3DS Outscale. Microsoft is strong in enterprise and cloud, while Atos/Eviden and Thales have deep defense experience. IBM and Entrust offer PQC-specific products, and ID Quantique specializes in quantum-safe solutions. Airbus Protect brings aerospace defense expertise. The customer will compare Orange's sovereign, certified approach against these specialized players.

Sources: SIG-5C7568EBD975 Elsevier BV 2026-01-01

Competitor	Category	Position here	Basis	Evidence
Microsoft	Hyperscaler	named in this topic's evidence — also an Orange partner	evidenced	SIG-5C7568EBD975
Atos / Eviden	Global systems integrator	sells Post-quantum cryptography; active in Defense; competes in Cybersecurity	structural	—
IBM	Global systems integrator	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Airbus Protect	Security specialist	sells Post-quantum cryptography; active in Defense; competes in Cybersecurity	structural	—
Entrust	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
ID Quantique	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Thales	Security specialist	sells Post-quantum cryptography; active in Defense; competes in Cybersecurity	structural	—
Dassault Systemes / 3DS Outscale	Sovereign cloud provider	active in Defense	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. What is your current cryptographic asset inventory, and have you identified which systems use RSA or ECC that would be vulnerable to quantum attacks?
2. Have you received any compliance mandates (e.g., CNSA 2.0) that set a deadline for PQC migration? If so, what is the timeline?
3. Do you have any systems that handle data with a long confidentiality requirement (e.g., 10+ years) that would be at risk from harvest-now-decrypt-later?
4. How are you currently managing cryptographic keys and certificates across your embedded and OT systems?
5. What is your budget and governance structure for a multi-year migration project?
6. Have you already engaged with any PQC vendors or consultants? If so, what did you learn?

Objections you will meet

They will say	You can answer
We don't see an immediate threat; quantum computers are years away.	That's true, but the harvest-now-decrypt-later risk means adversaries can record encrypted traffic today and decrypt it later. For defense systems with long data lifetimes, that's an immediate concern.
We already have a cybersecurity team; why do we need Orange?	Your team is likely stretched, and PQC migration is a multi-domain, multi-year effort. Orange brings specialized defense and cybersecurity expertise, plus sovereign cloud certifications that are critical for national security systems.
We've heard that PQC migration is complex and expensive; we're not sure it's worth it.	It is complex, but the cost of inaction could be much higher if your encrypted data is compromised. Orange can help you prioritize and phase the migration to manage costs.

Risks and unknowns

The main risk is that PQC migration is a multi-year effort with high complexity, and customers may delay due to cost or lack of urgency. There is also uncertainty about the exact timeline for a cryptographically relevant quantum computer, which could affect prioritization. Additionally, Orange's PQC offerings are not as mature as some competitors, and the lack of a dedicated PQC product line may be a disadvantage. The migration of embedded systems is particularly challenging due to limited tooling.

Next action, by role (FR-17)

Role	Do this next

Presales / Proposal	Prepare a bid brief for a defense agency's PQC migration project, highlighting Orange's ISO 27001 and SecNumCloud certifications and the Quantum Defender / quantum-safe offers, and include a narrative on the challenges of PKI migration and hybrid certificate designs.
Sales	In a conversation with a defense prime contractor's CISO, say: 'Given the CNSA 2.0 deadlines and the harvest-now-decrypt-later threat, we should discuss how our Quantum Defender / quantum-safe offers can help you migrate your PKI and secure your systems.'
Strategist / Innovator	Commission a deep-dive brief on the CNSA 2.0 migration timelines and the current state of NIST FIPS 203/204/205 adoption in defense and government sectors, leveraging Orange Group researchers and defense experts to map the opportunity for Quantum Defender / quantum-safe offers.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-40B64E273856	2026-07-30	International Journal of ...	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-B5F1E5C72CAB	2026-07-30	Springer Science and Busi...	1	Post-Quantum Cryptography Migration Readiness in Global Capability Centres: A Governance Framew...
SIG-B8561992126B	2026-07-30	openalex.org	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-57F4F0DAE044	2026-06-18	Cryptography	1	Mythos-Class Frontier Models and the Compression of Post-Quantum Cryptography Migration Timelin...
SIG-81E56EEBF9D4	2026-06-15	openalex.org	1	Post-Quantum Cryptography Migration for Agentic AI Systems
SIG-C822609B6A07	2026-05-31	openalex.org	1	Large Scale Actualisation of Post-Quantum Cryptography Migration
SIG-F119CA024DC5	2026-05-27	openalex.org	1	Post-Quantum Cryptographic Migration Strategies for Enterprise Systems
SIG-444983B4C996	2026-05-09	openalex.org	1	Parameter-Resident Cryptographic Material as an Unscoped Surface for Post-Quantum Migration: An...
SIG-BFFA5CE48BF5	2026-02-27	Shanlax International Jou...	1	Migration Strategies from Traditional Cryptography to Post-Quantum Cryptography in Cloud Systems
SIG-13AAE2CD702E	2026-01-19	MDPI AG	1	Cryptographic Asset Discovery and Inventory for Embedded Systems: A Framework for Post-Quantum ...
SIG-53F3508BD517	2026-01-09	openalex.org	1	Post-Quantum Public Key Infrastructures: Hybrid Certificates, Cryptographic Combiners, and Migr...
SIG-CE5A95F7AF72	2026-01-06	openalex.org	1	PQC Planner: Post-Quantum Cryptographic Migration Planner
SIG-353070727CBB	2026-01-01	openalex.org	1	Institutional Approaches to Post-Quantum Cryptography: A Comparative Analysis of Migration Fram...
SIG-5C7568EBD975	2026-01-01	Elsevier BV	1	The Role of the Second Line of Defense in Post-Quantum Cryptography Migration and Cryptographic...
SIG-D81DF125FECB	2026-01-01	Proceedings of the 23rd I...	1	A Capability-Driven Framework for Prioritizing Post-Quantum Cryptography Migration in Enterpris...
SIG-8BD0B85D3181	2025-12-24	openalex.org	1	Enterprise Migration to Post-Quantum Cryptography: Timeline Analysis and Strategic Frameworks
SIG-CA65D28B8AC6	2025-12-18	openalex.org	1	Harvest-Now, Decrypt-Later: A Temporal Cybersecurity Risk in the Quantum Transition
SIG-ED58CE7F1766	2025-11-24	openalex.org	1	Estimating Migration Timelines for Enterprises Transitioning to Post-Quantum Cryptography: A Re...
SIG-7ECDD0A30D62	2026-08-13	techtimes.com	2	Space Encryption Hardware Gets Fivefold Boost : Northrop Grumman and Aeronix
SIG-47DFEDD3D7A2	2026-07-15	arxiv.org	3	Multivariate Cryptography-Based Anonymous Certificate Scheme

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:13+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:16+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:41:14+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.